

Meng (Chris) Qiu

CONTACT INFORMATION	University of California, Merced SSM 362A, 5200 North Lake Road, Merced, CA 95343 Email: mengqiu@ucmerced.edu Website: chrismengqiu.com	
POSITIONS	University of California, Merced Assistant Professor, Department of Psychological Sciences	2024 - present
EDUCATION	Ph.D. in Quantitative Psychology, University of Notre Dame <i>Advisor: Dr. Ke-Hai Yuan</i>	2018 - 2024
	M.S. in Applied and Computational Mathematics and Statistics, University of Notre Dame <i>Advisor: Dr. Lizhen Lin</i>	2019 - 2022
	M.A. in Measurement, Statistics and Evaluation, University of Maryland <i>Advisor: Dr. Gregory R. Hancock</i>	2014 - 2016
	B.S. in Agricultural and Resources Economics, University of Maryland	2012 - 2014
	B.A. in Regional Development in Rural Area, China Agricultural University	2010 - 2012
RESEARCH INTERESTS	Bayesian Nonparametrics, Mixture Modeling, Longitudinal Data Analysis, Missing Data, Machine Learning, Causal Inference	
PUBLICATIONS	<i>Methodology</i>	
([†] STUDENT MENTEE)	- Depaoli, S., Qiu, M., Liu, H., & Jauregui, M. (2026). Class selection in growth mixture models: Comparing information criteria to nonparametric and parametric Bayesian approaches. <i>Structural Equation Modeling: A Multidisciplinary Journal</i>, 33(1), 64–80. doi.org/10.1080/10705511.2025.2563187 Qiu, M., Paganin, S., Ohn, I., & Lin, L. (2025). Bayesian nonparametric latent class analysis with different item types. <i>Psychological Methods</i>. Advance online publication. doi.org/10.1037/met0000728 Qiu, M. & Yuan, K.-H. (2024). Label switching in latent class analysis: Correct assignment of individuals, accuracy of parameter estimates and confidence intervals. <i>Structural Equation Modeling: A Multidisciplinary Journal</i>, 31(2), 217–232. doi.org/10.1080/10705511.2023.2213842 Qiu, M., Paganin, S., Ohn, I., & Lin, L. (2023). Bayesian nonparametric latent class analysis for different item types. <i>Multivariate Behavioral Research</i>, 58(1), 156–157. doi.org/10.1080/00273171.2022.2160958 - Yu, M., Li, Y., & Qiu, M. (2019). Statistical inference of the relative concentration index for complex surveys. <i>Statistics in Medicine</i>, 38(21), 4083–4095. doi.org/10.1002/sim.8284	

Tutorial

1. **Qiu, M.** (2022) A tutorial on Bayesian latent class analysis using JAGS. *Journal of Behavioral Data Science*, 2(2), 127-155. doi.org/10.35566/jbds/v2n2/qiu

Application

3. Liu, Q., **Qiu, M.**, Nestor, B. A., Rodriguez, V. J., & Cole, D. A. (2024). Mixture model applications in depression phenotyping: practices, challenges, and recommendations. *Current Psychology*, 43(33), 27006-27020. doi.org/10.1007/s12144-024-06309-6
2. Zhao, W., Ding, F., & **Qiu, M.** (2024). Can the spatial structure of a metropolitan area bring regional economic benefits?—A case study of Melbourne. *Applied Economics*, 1-16. doi.org/10.1080/00036846.2024.2304656
1. Salamanca-Balen, N., **Qiu, M.**, & Merluzzi, T. V. (2021). COVID-19 pandemic stress, tolerance of uncertainty and well-being for persons with and without cancer. *Psychology & Health*, 38(10), 1402-1419. doi.org/10.1080/08870446.2021.2020273

MANUSCRIPTS

([†]STUDENT
MENTEE)

Under Revision or Review

5. **Qiu, M.** & Yuan, K.-H. (Revise and resubmit). What can bootstrap do in latent class analysis? A whole-process bootstrap framework. *Sociological Methods & Research*.
4. Jauregui, M., Liu, H., Depaoli, S., & **Qiu, M.** (Revise and resubmit). Bayesian growth mixture modeling with an unknown number of latent classes using reversible jump MCMC. *Psychological Methods*.
3. **Qiu, M.** & Si, Y. (Manuscript under review). Bayesian nonparametric multiple imputation for categorical, continuous, and mixed data: A tutorial using the R package **BNPImpute**.
2. **Qiu, M.**, Paganin, S., Ohn, I., & Lin, L. (Manuscript under review). Frequently asked questions about Dirichlet process mixture modeling.
1. Zhao, W., Ding, F., **Qiu, M.**, Nijkamp, P., & Kourtit, K. (Manuscript under review). Building smart and low-carbon cities: Transition paths in the AI era.

CONFERENCE PRESENTATIONS AND POSTERS

([†]STUDENT
MENTEE)

7. Geng, Y.[†], & **Qiu, M.** (2026, July). *Frequently asked questions about label switching in mixture modeling*. Poster presented at the 2026 International Meeting of the Psychometric Society, Yonsei University, Seoul, South Korea.
6. **Qiu, M.**, Jia, F., & Liu, R. (2026, June). *A Bayesian nonparametric multiple imputation method for handling missing data in item response theory*. Talk presented at the 2026 Modern Modeling Methods Conference, Fordham University, NY.
5. **Qiu, M.**, Paganin, S., Ohn, I., & Lin, L. (2022, October). *Bayesian nonparametric latent class analysis for different item types*. Poster presented at the 2022 Annual Society of Multivariate Experimental Psychology Conference, Monterey, CA.
4. **Qiu, M.** (2017, July). *A Bayesian zero-inflated Poisson regression analysis on campus security*. Poster presented at the 2017 International Meeting of the Psychometric Society, University of Zürich, Switzerland.

3. Yu, M., Li, Y., & **Qiu, M.** (2017, May). *Variance estimation of health disparity measures in complex survey settings*. Talk presented at the 2017 Modern Modeling Methods Conference, University of Connecticut, CT.
2. **Qiu, M.**, & Hancock, G. R. (2017, April). *Robustness of structural means model to non-normality*. Talk presented at the 2017 American Educational Research Association Annual Meeting, San Antonio, TX.
1. Man, K. W., Jiao, H., & **Qiu, M.** (2016, May). *Modeling directional local item dependence*. Talk presented at the 2016 Modern Modeling Methods, University of Connecticut, CT.

HOHORS AND AWARDS	Society of Multivariate Experimental Psychology (SMEP) Travel Award		2022
	The 2022 SMEP Annual Meeting		
	Psychology Department’s Excellence in Graduate Student Teaching Award		2021
	University of Notre Dame		
	Graduate Student Travel Award		2016
	University of Maryland		
	Ray A. Murray Scholarship Fund		2012 - 2013
	University of Maryland		
RESEARCH GRANTS	Academic Senate Faculty Research Grant. (2026 - 2027). University of California, Merced. “Generative AI for Monte Carlo Simulation Studies in the Social Sciences.” Principal Investigator, \$7,500.		
PROFESSIONAL EXPERIENCE	Department of Psychology, University of Notre Dame		
	Statistical Consultant		Jun 2022 - Aug 2022
	Avar Consulting, Rockvill, MD		
	Research Analyst		Jan 2016 - Jul 2018
	Joint Program in Survey Methodology, University of Maryland		
	Faculty Assistant		Jan 2016 - May 2017
TEACHING	University of California, Merced		
	PSY 290 Statistical Programming in R	Role: Course Instructor	Spring 2026
	PSY 010 Analysis of Psychological Data	Role: Course Instructor	Spring 2026
			Fall 2025
			Spring 2025
			Fall 2024
	University of Notre Dame		
	PSY 63199 Grad Seminar: Quantitative Study	Role: Course Instructor	Spring 2024
			Fall 2023
			Spring 2023
			Fall 2022
			Spring 2022

PSY 30100 Experimental Psychology I: Statistics	Role: Teaching Assistant	Fall 2021
		Spring 2021
		Fall 2020
		Spring 2020

MENTORING (* GRADUATED)	Ph.D. Students		
	Yanjia Geng	Role: Primary Advisor	2026 - present
	Madelin Jauregui	Role: Candidacy Project Committee Member	2025 - present
	Aleksandr Ivanov	Role: Master Thesis Committee Member	2024 - 2025
	Ihnwhi Heo*	Role: Dissertation Committee Member	2025 - 2026

ACADEMIC SERVICE	<i>University Service</i>		
	Department of Psychological Sciences, UC Merced		
	PSY191 Capstone Talk		Spring 2025
	PSY200B ProSEM Talk		Spring 2025
	<i>Professional Service</i>		
	Journal Reviewer: Psychometrika, British Journal of Mathematical and Statistical Psychology, British Journal of Psychology, Structural Equation Modeling, Journal of Behavioral Data Science, Methods in Psychology, American Educational Research Association		

LANGUAGE	English, Chinese
----------	------------------

REFERENCES	Ke-Hai Yuan , Professor, Department of Psychology, University of Notre Dame, kyuan@nd.edu	
	Zhiyong Zhang , Professor, Department of Psychology, University of Notre Dame, zzhang4@nd.edu	
	Lizhen Lin , Professor, Department of Mathematics, University of Maryland, lizhen01@umd.edu	